Searching Legislative Design Space for Compromise and Productivity

A Computational Framework for Comparing Congress-Style Institutional Designs

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Legislatures need enough productivity to act, but productivity alone can also mean volatility, agenda capture, or weakly supported lawmaking. This paper presents a reproducible computational framework for searching legislative design space rather than defending any single reform. The simulator routes shared synthetic legislatures and bill streams through Congress-style institutional bundles that vary agenda access, voting thresholds, alternative selection, citizen review, lobbying safeguards, affected-group protection, package bargaining, and reversibility.

The contribution is methodological. The simulator separates productivity from compromise, public alignment, weak public-mandate passage, proposer advantage, concentrated harm, lobbying capture, administrative cost, and correction over time. A single comparison campaign generates all reported tables and figures from one CSV artifact, with descriptive case half-ranges, seed checks, mechanism ablations, and manipulation-stress diagnostics. The current experiment is breadth-first: default enactment remains one burden-shifting stress test, while the main comparison spans conventional, deliberative, anti-capture, agenda-scarcity, policy-tournament, harm-protection, and review-oriented families. The paper argues for mechanism search: legislative designs should be compared as portfolios of incentives and safeguards, not as isolated passage thresholds.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing theory, concepts and paradigms**; • **Computing methodologies** → **Modeling and simulation**; • **Applied computing** → *Law*.

Additional Key Words and Phrases: legislative institutions, institutional design, computational simulation, agenda control, compromise, collective decision-making, voting systems

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1 Introduction

The motivating question is not whether one unusual voting rule should replace Congress. It is whether systematic design search can help identify legislative structures that better combine productivity, compromise, and representative responsiveness. A legislature can be unified without being democratic; authoritarian systems can appear cohesive because disagreement is suppressed. A useful representative system should instead help actors with competing interests reach compromises connected to public support, public benefit, and the burdens imposed on affected groups.

Default enactment, where a proposal passes unless a blocking coalition stops it, is one design in the search space, not the object of the project. It is useful as a stress test because it reverses the usual burden of coalition formation. The broader object is the mechanism bundle around any final yes-or-no vote: who may propose, how agenda access is rationed, whether information is improved before voting, whether opponents have scarce challenge rights, whether bill content can move toward compromise, whether concentrated losers receive protection, and whether bad laws can be corrected later.

This paper therefore makes a modest claim. It does not validate a new constitution and it does not forecast the U.S. Congress. It contributes an executable framework for comparing legislative mechanisms under common assumptions,

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with explicit metrics and reproducible artifacts. The current simulations are best read as comparative stress tests: they reveal when a rule produces output, when it produces compromise, and when those two diverge.

2 Related Work

The model draws on individual-level simulation, spatial voting, and institutional theory. Agent-based and computational political models are useful when heterogeneous actors interact under rules that create aggregate outcomes [de Marchi and Page 2014; Epstein 2006; Gilbert and Troitzsch 2005]. Spatial voting motivates one-dimensional ideal points and status-quo-relative utility [Downs 1957]. Legislative bargaining and agenda-setter models motivate explicit proposer identity and proposer gain [Baron and Ferejohn 1989; Romer and Rosenthal 1978]. Pivotal-politics and veto-player theories motivate comparing simple majority, supermajority, bicameral, and executive-veto systems [Krehbiel 1998; Tsebelis 2002]. Committee and agenda-control theory motivates treating floor denial, referral, information, and amendment control as outcomes rather than missing observations [Cox and McCubbins 2005; Gilligan and Krehbiel 1987; Krehbiel 1991; Shepsle 1979; Shepsle and Weingast 1987; Weingast and Marshall 1988].

Several tested mechanisms come from adjacent literatures. Lobbying is modeled as access, effort, information, public pressure, and participation bias rather than only direct vote buying [Austen-Smith 1993, 1995; de Figueiredo and Richter 2014; Gilens and Page 2014; Hall and Deardorff 2006; Hall and Wayman 1990]. Citizen-panel variants draw on deliberative polling and mini-public practice [Fishkin 2009; Organisation for Economic Co-operation and Development 2020]. Agenda-credit variants echo quadratic-voting and storable-vote work on expressing intensity under scarcity [Casella 2005; Lalley and Weyl 2018]. Reversibility variants draw on temporary legislation and sunset review [Gersen 2007; Ranchordas 2014]. Policy tournaments draw on Condorcet and agenda-manipulation problems [McKelvey 1976; Tideman 1987]. Model documentation follows ODD and ODD+D conventions [Grimm et al. 2010; Müller et al. 2013].

Default-effect rules have real analogues, although usually in bounded domains. WTO dispute-settlement reports are adopted unless there is appeal or consensus not to adopt [World Trade Organization 1994]. EU fiscal-governance enforcement has used reverse qualified-majority procedures [European Parliament and Council of the European Union 2011]. Tacit-acceptance and silence procedures allow decisions to take effect unless timely objection occurs [Churchill and Ulfstein 2000]. U.S. fast-track and disapproval structures similarly shift procedural burdens in limited contexts [Davis 2024; United States Congress 1996]. These analogues suggest the same design lesson as the simulator: burden-shifting rules need guardrails around proposal access, objection, information, and correction.

3 Simulator

Each run creates a synthetic legislature, bill stream, status quo, and institutional process. Legislators have ideal points in $[-1, 1]$, party labels, district preferences, and heterogeneous sensitivity to party pressure, constituency pressure, lobbying, reputation, and compromise. Bills have a proposer, policy position, public support, generated public benefit, lobby pressure, salience, private gain, issue domain, public-benefit uncertainty, affected-group support, and concentrated harm. Enacted bills update the status quo, so votes evaluate policy movement rather than isolated proposals.

The implementation separates voting rules from legislative processes. Voting rules interpret yes/no totals. Legislative processes wrap a bill in stages such as proposal access, leadership agenda control, committee information, committee gatekeeping, bicameralism, conference committees, presidential veto, judicial review, coalition confidence, citizen initiatives, agenda lotteries, quadratic attention, challenge vouchers, adaptive routing, norm erosion, sunset review, earned credits, proposal bonds, explicit budgeted lobbying, distributional-harm review, package bargaining, mediation, competing alternatives, citizen-panel review, public objection, and law registry review. This modularity is central: the same threshold behaves differently when proposals are open, filtered, challenged, revised, reviewed, or made reversible.

The stylized U.S.-like conventional benchmark uses House majority passage, a Senate 60 percent threshold, presidential veto, two-thirds override, and an agenda-access gate. The gate prevents the benchmark from treating every introduced bill as a floor bill; it schedules bills using observable proxies: public support, salience, policy stability, sponsor-party fit, partisan queue conflict, capture risk, concentrated harm, and uncertainty. It does not observe generated public benefit directly. This makes the conventional baseline pass the current calibration screen without claiming it is a fitted model of Congress.

4 Metrics and Campaign

The simulator reports a dashboard rather than a single welfare score. Higher-is-better metrics include productivity, enacted support, generated welfare, compromise, public alignment, legitimacy, anti-lobbying success, citizen certification, district alignment, and welfare per submitted bill. Lower-is-better metrics include gridlock, chamber low-support passage, weak public-mandate passage, popular failure, lobbying capture, public-preference distortion, minority harm, concentrated-harm passage, proposer gain, administrative cost, low-support active-law share, and false-negative default passage. Floor consideration, access denial, committee rejection, challenge use, amendment use, compensation use, reversal, attention spending, objection-window use, and policy shift are diagnostic: high values can mean useful activity or institutional strain depending on context.

The metric types matter. Some metrics are structural outputs, such as productivity, enactment counts, floor load, review load, and administrative cost. Some are synthetic value proxies generated inside the model, such as public support, public benefit, legitimacy, weak public mandate, and affected-group harm. Others are diagnostics, such as amendment, challenge, objection, and reversal rates. Chamber low-support passage is mainly a threshold-failure metric; under ordinary majority voting it is often mechanically low. Weak public-mandate passage is the cross-system public-support failure metric because it asks whether enacted bills had generated public support below 50 percent, regardless of the chamber vote rule.

For display, the paper reports a directional score only as a diagnostic reading aid. Representative quality averages generated welfare, enacted support, compromise, public alignment, and legitimacy. Risk control inverts chamber low-support passage, weak public-mandate passage, minority harm, lobbying capture, public-preference distortion, concentrated-harm passage, proposer gain, and policy shift. Administrative feasibility inverts estimated procedural load from attention spending, review, challenge, amendment, and alternative selection. The directional score averages productivity, representative quality, risk control, and administrative feasibility, but the analysis emphasizes Pareto tradeoffs because different value profiles favor different institutional designs.

All main results come from one reported comparison campaign. The campaign combines broad assumption cases, adversarial generator cases, weighted party-system sensitivity cases, and a stylized rising-contention timeline in one CSV. Broad cases vary polarization, party loyalty, compromise culture, lobbying pressure, constituency pressure, party count, proposal pressure, capture pressure, and public appetite for anti-lobbying reform. Adversarial cases test high-benefit extreme reform, popular harmful bills, moderate capture, delayed-benefit reform, concentrated rights-like harm, and anti-lobbying backlash. Party-system cases compare two-party, two-major-plus-minor, fragmented multiparty, and dominant-party profiles. Timeline cases increase polarization, party loyalty, lobbying, and proposal pressure while reducing compromise culture and constituency responsiveness.

Scenario selection follows a fixed coverage rule: include conventional baselines, the stylized U.S.-like benchmark, and one readable family representative for each implemented non-default design class, plus three default-enactment stress tests. The paper build writes `reports/scenario-selection-manifest.md`, which records each included row's family role and selection rationale. Non-default representatives include leadership agenda control, regular order, cloture with conference and review, coalition confidence, direct initiative, policy tournaments, citizen review, agenda scarcity, proposal accountability, affected-group protection, package bargaining, anti-capture safeguards, public objection, risk routing, norm erosion, law review, and one synthesized portfolio hybrid. The hybrid was added after comparing the first campaign results: it gives low-risk bills a fast lane, sends middle-risk bills through pairwise alternatives and mediation, sends high-risk bills through citizen and harm review, and wraps the system in proposer bonds, lobbying safeguards, and law-registry correction. When a family has many implemented variants, the paper uses the variant that best represents the family mechanism rather than the variant with the highest observed score. A supplemental `make family-screen` run evaluates every explicit scenario key and reports the highest-scoring scenario within each family under a fixed baseline rule; stressed policy-tournament variants with clone and decoy flooding remain there rather than being counted as a second main policy-tournament row. Earlier generated reports keep deeper default-pass sweeps as a side log, but the main comparison campaign treats default enactment as one comparison family rather than the organizing case.

The simulator now includes a separate chamber-structure supplement for seat allocation, chamber routing, committee assignment, committee power, eligibility/appointment filters, and independent ex ante review. This supplement is not merged into the main evidence table because it asks a different design question: how representation architecture and intermediate institutions change the same productivity/compromise tradeoff. Its scenarios vary equal-population and proportional chambers, malapportioned and appointed upper chambers, origination rules, upper-house amendment or veto powers, navette, mediation, last-offer bargaining, lower-house override, committee selection rules, mixed citizen committees, public-accounts and legal-review committees, recusal and cooling-off rules, and insulated fiscal/electoral/audit review bodies.

Calibration is a screening step, not a parameter fit. `make calibrate` compares conventional baselines with broad benchmark ranges derived from Voteview, Congress.gov, govinfo, Comparative Agendas Project, ParlGov, lobbying-disclosure and OpenFEC records, Center for Effective Lawmaking measures, and V-Dem indicators [Comparative Agendas Project 2026; Döring and Manow 2024; Federal Election Commission 2026; Lewis et al. 2021; Library of Congress 2026; United States Government Publishing Office 2020; United States Senate Office of Public Records 2026; V-Dem Institute 2026; Volden and Wiseman 2014]. These are proxy checks for gross plausibility, not validation against raw institutional data. The current screen passes all tracked checks, including the stylized U.S.-like benchmark's enactment rate and floor load.

5 Results

Table 3 compares representative family examples across broad and adversarial assumption cases using the highest-signal columns; the appendix reports the full scenario table with directional and seed diagnostics. Table 4 makes the narrow productivity/compromise/risk frontier explicit. The frontier should not be read as a recommendation: it excludes administrative feasibility and treats generated risk control as a single combined objective. Table 5 adds a second diagnostic layer: ablations ask which components appear to do work, while stress rows ask how promising mechanisms degrade under manipulation pressure.

Table 1. Compact design-space index for the main comparison campaign. Labels match Table 3 and the scatter plots; detailed mechanism descriptions are in the appendix and supplement.

Family	Labels	Mechanism theme	Primary stress risk
Conventional	SM S60 BIC VETO CUR	Thresholds/vetoes	Bottlenecks
Procedure	LEAD PROC COMM	Agenda control	Queue capture
Courts	COURT	Review architecture	Juridification
Public model	DIST	District publics	Proxy opinion
Agenda gates	SCR LOT	Screen/lottery	Gate bias
Coalition	PARL	Confidence/access	Bloc cartels
Direct democracy	INIT	Initiative path	Campaign bias
Tournaments	PAIR	Alternatives	Clones/decoys
Mini-public	JURY	Citizen burden shift	Manipulation
Scarcity	QAB BOND	Credits/bonds	Hoarding
Harm rules	HARM COMP	Minority protection	Holdouts
Bargaining	PKG MPKG OMNI	Package trades	Proxy limits
Correction	LAW OBJ	Review/objection	Instability
Anti-capture	CAP INFL	Lobby safeguards	Evasion
Adaptive	RISK NORM LEARN	Risk/norm/learning	Gaming
Portfolio	PORT XPORT	Mechanism bundle	Complexity
Default stress	DP DPC DPM	Burden shift	False silence
Out of scope	—	Elections/media/etc.	Endogeneity

Table 2. Chamber, apportionment, committee, and review family champions from the chamber-structure supplement. Values are averaged across the chamber campaign cases.

Label	Family champion	Prod.	Comp.	Risk	Malapp.
BICAM	Bicameral conflict	0.31	0.27	0.86	0.00
ASSIGN	Committee assignment	0.22	0.29	0.89	0.00
COMMP	Committee power	0.25	0.28	0.87	0.00
CONV	Conventional benchmark	0.34	0.25	0.85	0.00
REVIEW	Independent review	0.34	0.25	0.85	0.00
ROUTE	Origination and routing	0.31	0.27	0.86	0.00
OTHER	Other chamber design	0.31	0.26	0.86	0.00
SEAT	Seat allocation	0.34	0.25	0.85	0.00
SELECT	Selection and retention	0.36	0.24	0.84	0.00
UPPER	Upper-chamber composition	0.36	0.26	0.84	0.12

The main results are easiest to read as tradeoffs rather than rankings. Pairwise alternatives are the strongest non-default example in the reported campaign: they increase productivity and compromise while sharply reducing weak public-mandate passage, but the ablation and stress diagnostics show that they are procedurally costly and vulnerable to clone or decoy pressure. The synthesized portfolio hybrid performs as a plausible next design hypothesis rather than as a clean winner: it gives up some pairwise-tournament compromise and throughput, but it keeps most of the productivity gain while improving capture and risk-control diagnostics through proposal bonds, public-interest access, citizen/harm review, lobbying safeguards, and later law review. Open default enactment has the highest throughput and a high diagnostic score, but it also has the largest weak public-mandate burden and poor risk control. Multidimensional package bargaining currently does not improve the displayed value profile: the added cross-issue trade machinery raises administrative cost faster than it improves compromise. The stylized U.S.-like benchmark is conservative: it has low productivity, low weak public-mandate passage, and high administrative feasibility. In this synthetic model, conventional-style gatekeeping can also protect generated welfare by allowing few bills through; that is not a dominance result because the same gate imposes a severe productivity cost.

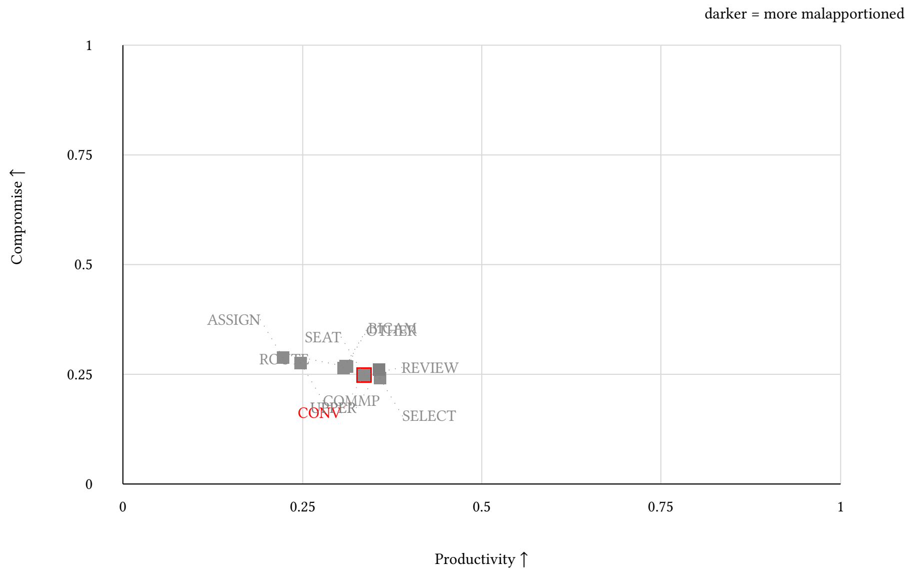


Fig. 1. Chamber-structure family champions from the supplemental chamber campaign. The axes keep the paper's central comparison visible, while grayscale intensity marks the generated malapportionment index. The figure is a structural sensitivity view, not a replacement for the main scenario comparison.

Table 3. Representative family comparison from the main campaign. Metric entries are mean±case half-range across broad and adversarial cases. The full table with directional and seed diagnostics is in the appendix and supplement.

Label	Representative system	Prod. ↑	Comp. ↑	Weak mand. ↓	Risk ctrl. ↑	Admin ↓
CUR	**Stylized U.S.-like conventional benchmark**	**0.052±0.103**	**0.368±0.184**	**0.007±0.022**	**0.929±0.045**	**0.016±0.014**
SM	Unicameral simple majority	0.320±0.458	0.245±0.193	0.213±0.291	0.852±0.087	0.070±0.000
COMM	Committee-first regular order	0.207±0.409	0.287±0.184	0.098±0.119	0.891±0.099	0.032±0.022
PARL	Parliamentary coalition confidence	0.225±0.425	0.292±0.267	0.067±0.194	0.904±0.115	0.019±0.030
INIT	Citizen initiative and referendum	0.430±0.385	0.221±0.209	0.223±0.278	0.833±0.072	0.070±0.000
PAIR	Unicameral majority + pairwise alternatives	0.542±0.207	0.502±0.093	0.002±0.003	0.937±0.067	0.285±0.044
JURY	Citizen assembly threshold gate	0.197±0.175	0.283±0.216	0.061±0.227	0.907±0.077	0.190±0.000
QAB	Quadratic attention budget + majority	0.247±0.366	0.265±0.189	0.178±0.292	0.878±0.076	0.220±0.033
BOND	Proposal bonds + majority	0.319±0.455	0.245±0.195	0.213±0.290	0.852±0.084	0.070±0.002
HARM	Harm-weighted double majority	0.271±0.449	0.261±0.196	0.202±0.286	0.876±0.069	0.070±0.000
MPKG	Multidimensional package bargaining + majority	0.341±0.443	0.240±0.199	0.218±0.205	0.835±0.081	0.173±0.116
CAP	Majority + anti-capture bundle	0.306±0.197	0.247±0.177	0.152±0.226	0.884±0.056	0.065±0.016
RISK	Risk-routed majority legislature	0.530±0.331	0.264±0.170	0.231±0.171	0.843±0.070	0.171±0.029
PORT	Portfolio hybrid legislature	0.503±0.303	0.421±0.063	0.030±0.026	0.933±0.034	0.240±0.096
XPORT	Expanded portfolio hybrid legislature	0.784±0.239	0.463±0.052	0.002±0.004	0.905±0.052	0.607±0.107
LAW	Active-law registry + majority review	0.346±0.463	0.236±0.175	0.244±0.301	0.835±0.090	0.118±0.048
DP	Default pass unless 2/3 block	0.866±0.089	0.136±0.221	0.472±0.370	0.630±0.065	0.070±0.000
DPM	Default pass + multi-round mediation + challenge	0.802±0.177	0.243±0.171	0.374±0.226	0.722±0.070	0.206±0.042

Note: Case half-ranges are descriptive variation across campaign cases, not statistical confidence intervals. Red marks the stylized U.S.-like conventional benchmark.

The chamber-structure supplement adds a second, flatter lesson. Table 2 and Figure 1 show that chamber architecture changes the tradeoff surface, but no structural family dominates the mechanism-level campaign. Selection/retention filters and independent-review bundles have the highest chamber-family directional diagnostics; malapportioned

Table 4. Pareto-front scenarios under productivity, compromise, and risk control.

Label	Role	Prod. ↑	Comp. ↑	Risk ↑	Weak mand. ↓	Admin ↓
DP	Throughput endpoint	0.866	0.136	0.630	0.472	0.070
DPM	Throughput with mediation/challenge	0.802	0.243	0.722	0.374	0.207
XPORT	Non-dominated profile	0.784	0.463	0.905	0.002	0.607
PAIR	Alternative-comparison endpoint	0.542	0.502	0.937	0.002	0.285

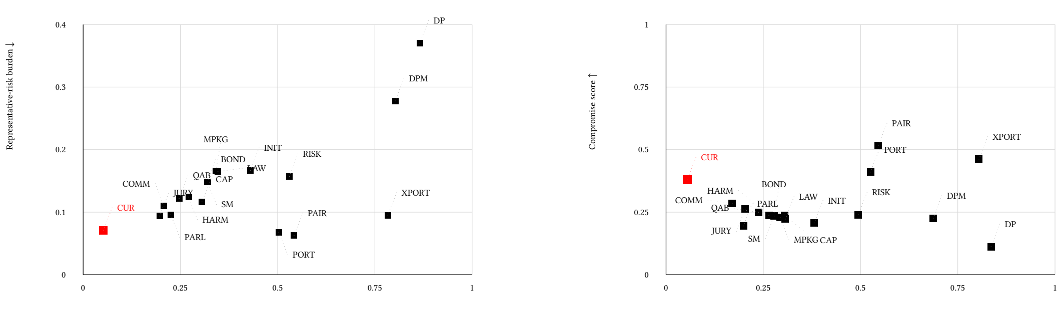
Note: The frontier changes if administrative feasibility, welfare, or rights/harm priorities are hard objectives.

Table 5. Ablation and manipulation-stress diagnostics for selected mechanisms.

Type	Mechanism	Comparison	Dir.	Comp.	Weak mand.	Admin
Ablation	Policy tournament	No alternatives	+0.046	+0.262	+0.206	+0.215
Ablation	Risk routing	No jury lane	-0.002	+0.000	+0.001	+0.009
Ablation	Anti-capture	Screen only	+0.019	+0.000	+0.028	+0.005
Ablation	Package bargaining	Scalar package	-0.015	-0.002	+0.002	+0.063
Ablation	Reversibility	No registry	-0.011	-0.009	-0.030	+0.049
Stress	Tournament attack	Clones/decoys	+0.069	+0.046	+0.010	-0.077
Stress	Citizen-panel manipulation	Noisy panel	+0.001	+0.018	-0.005	+0.000
Stress	Bad-faith harm claims	Loose claims	-0.003	-0.018	-0.018	+0.000
Stress	Objection astroturf	Astroturf/noise	+0.011	-0.006	+0.003	+0.010
Stress	Agenda flooding	Proposal flood	+0.006	-0.011	+0.007	-0.001

Note: Ablation weak mandate is a reduction, so positive is better; admin is added cost, so positive is worse. For stress rows, positives are losses or added burden.

upper-chamber designs can raise productivity in some stressed cases, but the supplement exposes the representational cost through the malapportionment and weak-mandate columns. Committee-first and legal/drafting committees do most of their work as risk filters, while origination and bicameral conflict rules mostly shift delay and veto burdens.



(a) Representative-risk burden

(b) Compromise score

Fig. 2. Two productivity tradeoff views for the representative systems in Table 3, rendered side by side with the same 0–1 productivity axis. Panel (a) plots representative-risk burden, defined as 1−risk control; lower is better. Panel (b) plots compromise under weighted party-system sensitivity cases; higher is better. The stylized U.S.-like benchmark is red and marked with a larger square.

Label	Productivity	Compromise	Rep. quality	Risk ctrl.	Admin feas.
CUR	**0.05**	**0.37**	**0.65**	**0.93**	**0.98**
SM	0.32	0.25	0.56	0.85	0.93
COMM	0.21	0.29	0.59	0.89	0.97
PARL	0.23	0.29	0.58	0.90	0.98
INIT	0.43	0.22	0.58	0.83	0.93
PAIR	0.54	0.50	0.65	0.94	0.71
JURY	0.20	0.28	0.61	0.91	0.81
QAB	0.25	0.26	0.57	0.88	0.78
BOND	0.32	0.25	0.56	0.85	0.93
HARM	0.27	0.26	0.57	0.88	0.93
MPKG	0.34	0.24	0.56	0.83	0.83
CAP	0.31	0.25	0.58	0.88	0.93
RISK	0.53	0.26	0.54	0.84	0.83
PORT	0.50	0.42	0.62	0.93	0.76
XPORT	0.78	0.46	0.62	0.90	0.39
LAW	0.35	0.24	0.55	0.84	0.88
DP	0.87	0.14	0.48	0.63	0.93
DPM	0.80	0.24	0.51	0.72	0.79

Representative Table rows; darker cells are higher/better-oriented values

Fig. 3. Value-profile matrix for selected systems. Rows follow the labels in Table 3, and every cell is oriented so darker/larger values are better under the stated metric interpretation. This figure should be read as a profile comparison, not as a single ranking.

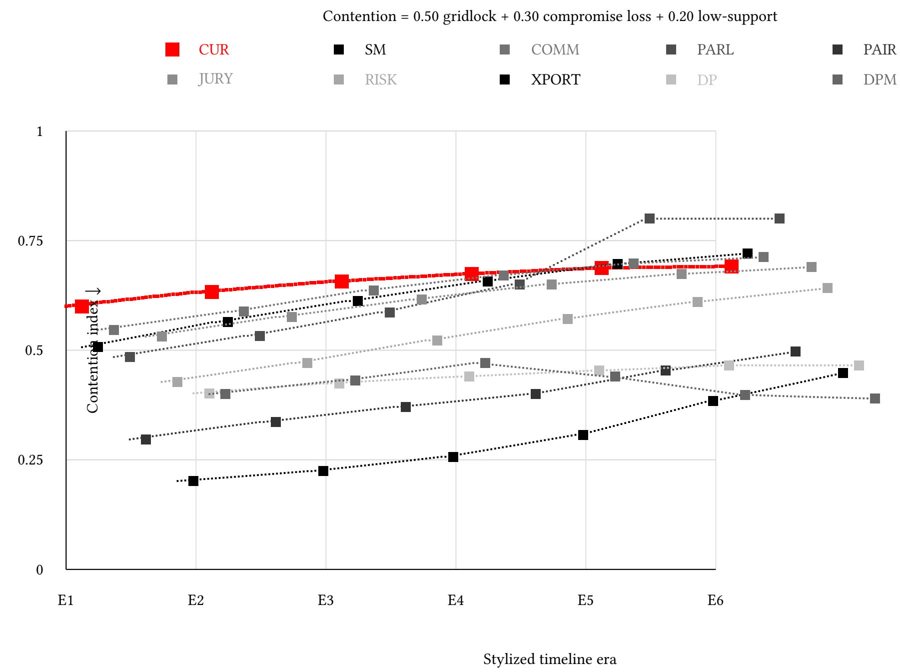


Fig. 4. Rising-contention stress path from the main comparison campaign. The index combines gridlock, compromise loss, and weak public-mandate passage. Lower is better. This is a parameterized stress path, not an endogenous causal model of polarization.

Three patterns are observed in this exploratory campaign. First, passage thresholds alone are a weak design language. Supermajority and conventional baselines suppress weakly mandated enactment, but they also suppress throughput. Open default enactment solves throughput but creates risk. Second, agenda and content mechanisms matter across rule families, not only inside default pass. Alternative-comparison mechanisms are promising because they reduce proposer agenda advantage, but stressed clone and decoy variants remain a warning against treating tournaments as automatically robust. Leadership, committee-first, and public-interest gates reduce weak bills at high access cost; citizen review, direct initiatives, and public objection change burdens of proof; coalition-confidence and attention-budget systems ration scarce agenda access; and harm, package-bargaining, and compensation rules target legitimacy rather than raw throughput. Third, reversibility is a distinct tradeoff. Judicial review and law-registry review reduce exposure to weak or rights-risky laws but can increase status-quo volatility because correction itself changes policy.

The exploratory result therefore does not support a single dominant passage rule or one universal guardrail. The explicit portfolio-hybrid scenario turns the observed pattern into a testable design: preserve a fast lane for low-risk proposals, require stronger consent or revision for risky proposals, make proposers accountable for low-quality bills, expose organized-interest pressure, compare alternatives before binary ratification, protect affected groups, and review enacted laws when uncertainty is high. The expanded portfolio variant adds district-public signals, long-horizon proposer adaptation, omnibus-style package bargaining, campaign-finance influence safeguards, and constitutional-court architecture. Its mixed result is useful: combining good mechanisms improves several risk diagnostics, but it also adds administrative load and does not automatically dominate simpler pairwise-alternative designs.

6 Limitations

The model is deliberately stylized. It uses one primary policy dimension, generated legislators, generated public support, generated public benefit, simplified issue domains, and abstract lobbying units. Public benefit is internal to the generator, so welfare comparisons are conditional on model assumptions. This creates a circularity risk: systems that screen extreme, uncertain, or lobby-heavy bills may look better partly because the generator often encodes those features as lower expected public value. Adversarial generator cases partially relax this assumption, but the paper still treats welfare as a shared synthetic signal, not as an empirical estimate of social welfare.

The calibration layer screens conventional baselines against named empirical ranges, but it does not fit raw datasets. The repository includes optional adapters for bill histories, roll-call coalitions, topic throughput, lobbying distributions, district public opinion, committee activity, campaign finance, court review, and comparative institutions. Small Congress.gov and OpenFEC samples now exercise some adapters, but they are smoke tests, not validation data. Those adapters still need larger curated source extracts and tighter tolerances before supporting claims about real Congress. Strategic behavior also remains bounded. Proposers and lobby groups adapt, and norm erosion can raise procedural delay, but parties, committees, coalitions, courts, agencies, media, and elections do not yet co-evolve over long horizons. Chamber-structure scenarios are especially stylized: apportionment, independence, appointment, committee expertise, and review-body insulation are modeled as sensitivity parameters, not calibrated constitutional facts. Citizen panels, initiatives, objection windows, tournaments, audits, challenge tokens, judicial review, and law-registry review are optimistic mechanism prototypes: their transaction costs, manipulation risks, and administrative burdens are represented only indirectly through diagnostics such as attention spending, amendment load, objection use, review use, and volatility. Omnibus and package-bargaining modules add side payments, delay costs, jurisdictional trades, and distributive-load proxies, but they still do not model actual legal text, appropriations, or high-dimensional coalition bargaining.

7 Reproducibility

The submitted manuscript is anonymized for double-blind review. The reproducibility package contains source code, generated campaign reports, calibration reports, the ODD+D appendix, and scripts. The main results are regenerated with:

```
make paper-campaign
make calibrate
make empirical-bridge
make ablation-analysis
make manipulation-stress
make seed-robustness-check
make family-screen
make catalog-breadth
make chamber-structure
make validation-readiness
make empirical-validation
make paper
make supplement-anonymous
```

The paper build regenerates the main comparison campaign before drawing tables or figures, so the PDF is tied to the tracked campaign artifact rather than to stale CSV files. The seed-robustness target writes a multi-seed report for every paper scenario, and the full appendix table includes a seed half-range column for the directional diagnostic alongside case half-ranges across campaign cases. The ablation and manipulation-stress targets write diagnostic reports for component removal and strategic attack cases. The empirical-bridge target maps optional raw-data summaries to the calibration screen, making clear which real-world signals are still missing. The family-screen target writes a supplemental breadth-catalog comparison under a fixed selection rule; catalog-breadth records which historical default-pass variants are archived outside that screen; and the generated selection manifest documents why each paper scenario is included. The anonymous supplement target packages source, paper files, generated reports, and review PDFs while excluding repository metadata and local artifacts. A public repository URL will be added after review if anonymity rules permit; a separate public provenance manifest can be generated after review with `make public-provenance`. Optional public API smoke-test samples can be refreshed with `make fetch-validation-samples` when Congress.gov and OpenFEC keys are available.

Use of Large Language Model Tools

The research question and core simulator idea originated with the human author. Large language model tools were used for software implementation assistance and for drafting, editing, and reorganizing manuscript text. The author reviewed and is responsible for all code, experiments, reported results, text, and citations. No large language model is listed as an author.

8 Conclusion

This paper argues for computational mechanism search as a way to study legislative design. Default enactment is one useful stress case, but the broader contribution is the ability to compare ordinary and unusual institutional bundles under common assumptions. The current prototype suggests that compromise and productivity are most likely to improve through combinations of agenda access, contestation, mediation, alternative selection, anti-capture safeguards, affected-group protection, and reversibility rather than through any single passage threshold or one favored procedural family. The portfolio and expanded-portfolio hybrids are synthesized candidates produced by that comparison; the next scientific step is to fit and stress-test their public-opinion, influence-system, court-review, and bargaining submodels rather than treating either as a final answer.

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